

## 1-NETWORK BASICS

### Types of Network:

#### ① LAN [Local Area Network]

\* Network within a small area. (home, office, school)

#### ② WAN [wide Area Network]

\* Network covering large geographical areas (Internet).

#### ③ MAN [Metropolitan Area Network]

\* Network covering a city or metropolitan area.

### ③ PAN [Personal Area Network]

\* Network around a person (Bluetooth, Mobile hotspot)

### ④ CAN [Campus Area Network]

\* Network connecting Multiple buildings in a Campus or University.

### ⑤ SAN [Storage Area Network]

\* Dedicated high-speed network for connecting Storage devices and Servers.

### ⑥ MANET [Mobile Ad hoc Network]

\* Wireless network formed directly between mobile devices without a Router.

## Network Models :

### ① Client - Server Model

\* Clients request Services

\* Servers provide Services.

② peer-to-peer (P2P) networks:

All devices can share resources directly without a Central Server

Ex: Torrent File sharing.

③ Hybrid Model:

+ Combination of client-server and p2p.

Ex: Many Modern Applications.

④ Centralized Networks:

+ One central device/server controls everything.

⑤ Distributed Networks:

Multiple systems share workload and data.

Internet vs Intranet vs Extranet:

① Internet:

Public global network accessible by everyone.

② Intranet:

Private network used within an organization.

③ Extranet:

private network that allows limited access to external Users (partners/customers).

## 2. DATA COMMUNICATION BASICS

① Bandwidth

Maximum amount of data that can be transmitted over a network per second.

Ex:

A 100 Mbps internet connection can carry up to 100 megabits per second.

② Throughput

Actual Amount of data Successfully transferred : per second.

Ex:

you have 100 Mbps bandwidth, but due to congestion you only get 80 Mbps.

- Bandwidth = 100 Mbps
- Throughput = 80 Mbps.

### ③ Latency

Time taken for data to travel from source to destination.

Ex:

you ping Google and get 20 ms.

† This means the packet took 20 milliseconds to reach and return.

### ④ Jitter

Variation in latency between packets.

Ex:

packet 1 = 20 ms

packet 2 = 25 ms

packet 3 = 18 ms

Latency keeps changing  $\rightarrow$  Jitter exists.

### ⑤ packet loss

Packets that never reach the destination.

Ex:

100 packets sent.

95 packets received.

5 packets lost.

} packet loss = 5%



## ⑥ MTU [Maximum Transmission Unit]

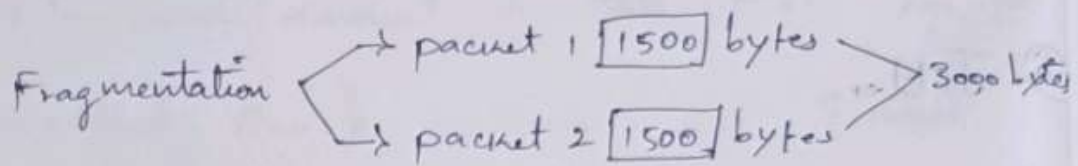
Maximum packet size that can be sent without fragmentation.

Ex:

Ethernet MTU = 1500 bytes.

If data = 3000 bytes.

→ Split it into 2 packets.



## Communication Modes

### ⑦ Simplex

Data flows in only one direction.

Ex:

\* TV Broadcast

\* Sender → Receiver

\* Radio

\* No reply possible.

### ⑧ Half Duplex

Data can flow both ways, but not at the same time.

Ex:

Walkie - Talkie

x person A talks  $\rightarrow$  person B listens

x person B talks  $\rightarrow$  person A listens.

\* Not Simultaneously.

### ⑨ Full Duplex

Data flows both ways Simultaneously.

Ex:  
phone call.

\* Both people can speak and listen at the same time.

## 3. NETWORK DEVICES

### ① Hub

Connects devices and sends data to all ports.

Ex:

PC 1 sends data  $\rightarrow$  Hub broadcasts it to every connected device.

## ② Switch

Connects devices and forwards data Only to the correct device using MAC addresses.

Ex:

PC1 Sends data to PC2 → Switch Sends  
Only to PC2

## ③ Router

Connects different networks and forwards data using IP addresses.

Ex:

Your home network connects to the Internet through a router.

## ④ Bridge

Connects two LAN segments and filters traffic using MAC addresses.

Ex:

Connects two office LANs into one larger network.



### ⑤ Repeater

Regenerates weak signals to extend network distance.

Ex:

wifi extender in a large house.

### ⑥ Gateway

Connects networks using different protocols.

Ex:

A network communicating with a completely different network type.

### ⑦ Modem

Converts ISP signals into digital signals for your network.

Modulator → DeModulator

Ex:

= Modem.

Fibre/cable internet enters the modem before reaching the router.

\* Modem = Internet Converter.

### ⑧ Wireless Access point (AP)

provides wireless Connectivity to devices.

Ex:

Laptop Connects to wi-fi through an AP.

### ⑨ Firewall

Monitors and filters network traffic based on Security Rules.

Ex:

Blocks unauthorized Connections from the Internet

### ⑩ Load Balancer.

Distributes traffic across multiple Servers.

Ex:

website traffic Shared among Several web Servers.

### ⑪ proxy Server

Acts as an intermediary between Users and Servers.

Ex:

User  $\rightarrow$  proxy  $\rightarrow$  Website.

### ⑫ IDS [Intrusion Detection System]

Detects Suspicious activity and generate alerts.

Ex:

Alerts Security team about possible attacks.

IDS = Alarm System.

### ⑬ IPS [Intrusion Prevention System]

Detects and automatically blocks malicious traffic.

Ex:

IPS = Alarm + Security Guard.

### ⑭ CSU/DSU

Connect Routers to leased lines (WAN Connection)

CSU/DSU = WAN Connector.

### ⑮ Network Interface Card (NIC)

Hardware that allows a device to connect to a network.

Ex:

Ethernet Card / wifi adapter in your laptop.

## ⑩ VPN Concentrator

Manages Multiple VPN Connections.

Ex:

Employees Securely Connect to the Company network remotely.

† VPN Concentrator = VPN Manager.